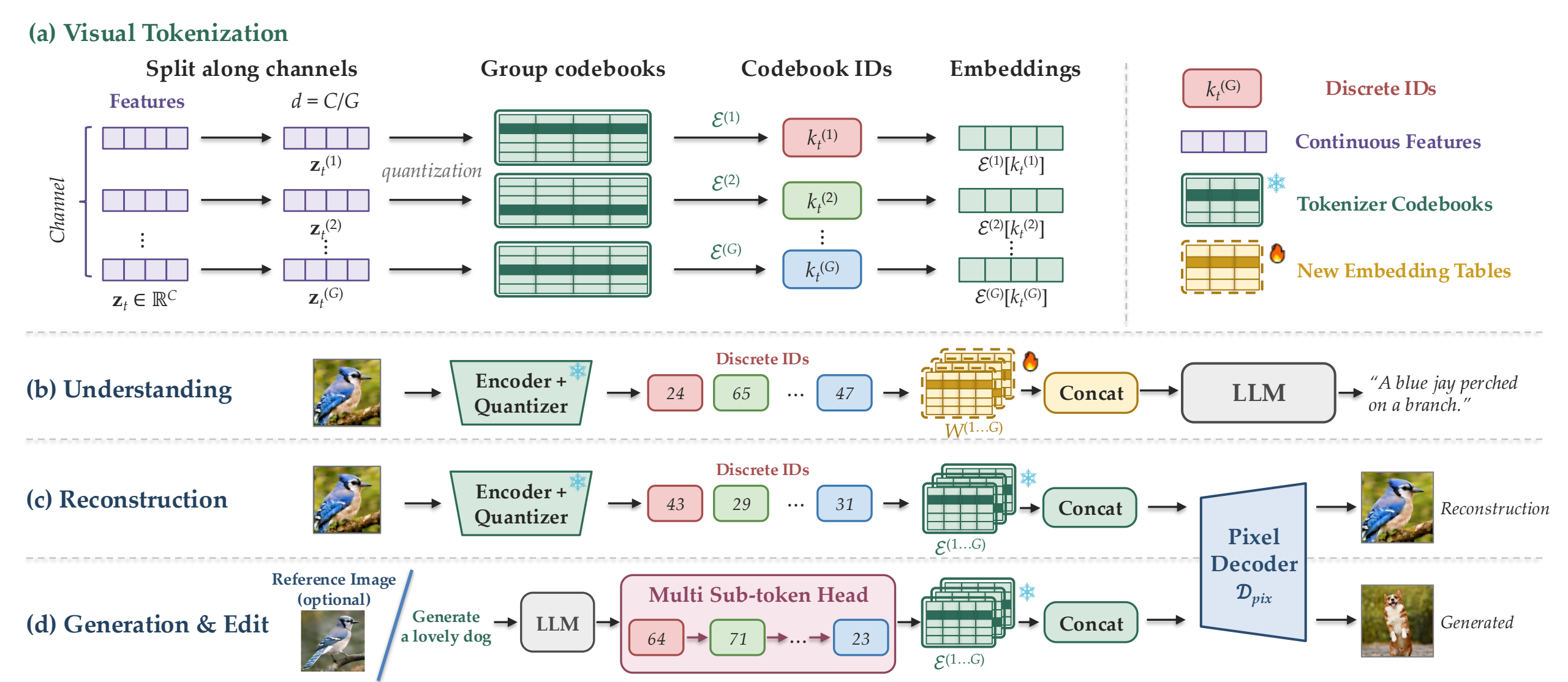

(a) Visual Tokenization
Split along channels
Group codebooks
Codebook IDs
Embeddings
Features
d = C/G
Channel
quantization
Discrete IDs
Continuous Features
Tokenizer Codebooks
New Embedding Tables
(b) Understanding
Encoder + Quantizer
Discrete IDs
24
65
47
Concat
LLM
"A blue jay perched on a branch."
(c) Reconstruction
Encoder + Quantizer
Discrete IDs
43
29
31
Concat
Pixel Decoder
Reconstruction
(d) Generation & Edit
Reference Image (optional)
Generate a lovely dog
LLM
Multi Sub-token Head
64
71
23
Concat
Pixel Decoder
Generated